

ZAP!

A high-level language that compiles to **6502 / 65C02 assembly**, targeting **Atari 8-bit** computers. Combines high-level constructs (structs, enums, functions, loops) with low-level control (pointers, hardware registers, inline assembly) and produces tight, optimized 6502 code.

[Get started](#) · [View on GitHub](#)

Quickstart

Save this as **hello.zap**:

```
.include "lib/atari/atari_stdio.zap"

proc main()
    puts("Hello, ZAP!")
end
```

Compile, assemble, and link:

```
zapc --cpu 6502 -O1 -I lib hello.zap -o hello.s
ca65 hello.s -o hello.o
ld65 -C cfg/my_atari.cfg hello.o -o hello.xex
```

The resulting **hello.xex** runs on any Atari 8-bit emulator (Altirra, Atari800) or real hardware.

Performance

In benchmark comparisons, ZAP! generates code that runs **~44% faster than Action!** on the Atari 8-bit platform.

Documentation

Browse the guides using the sidebar, or start with:

- [Getting Started Guide](#) — beginner tutorial
- [Language Reference](#) — complete language spec
- [Advanced Topics](#) — pointers, inline assembly, optimization
- [Standard Library](#) — built-in functions and modules
- [Error Messages](#) — compiler error catalog
- [Known Limitations](#) — current limitations
- [Architecture](#) — how the compiler works internally
- [IDE Integration](#) — VS Code setup

License

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